INCH-POUND

MIL-PRF-22885/103A
17 March 1999
SUPERSEDING
MIL-S-22885/103
15 October 1985

PERFORMANCE SPECIFICATION SHEET

SWITCHS, PUSHBUTTON, ILLUMINATED, 2 AMPERES, 2PDT, 4PDT AND INDICATOR, 4 LAMP, SUNLIGHT READABLE (EMI/RFI SHIELDING, MODULAR CONSTRUCTION)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification sheet and MIL-PRF-22885.

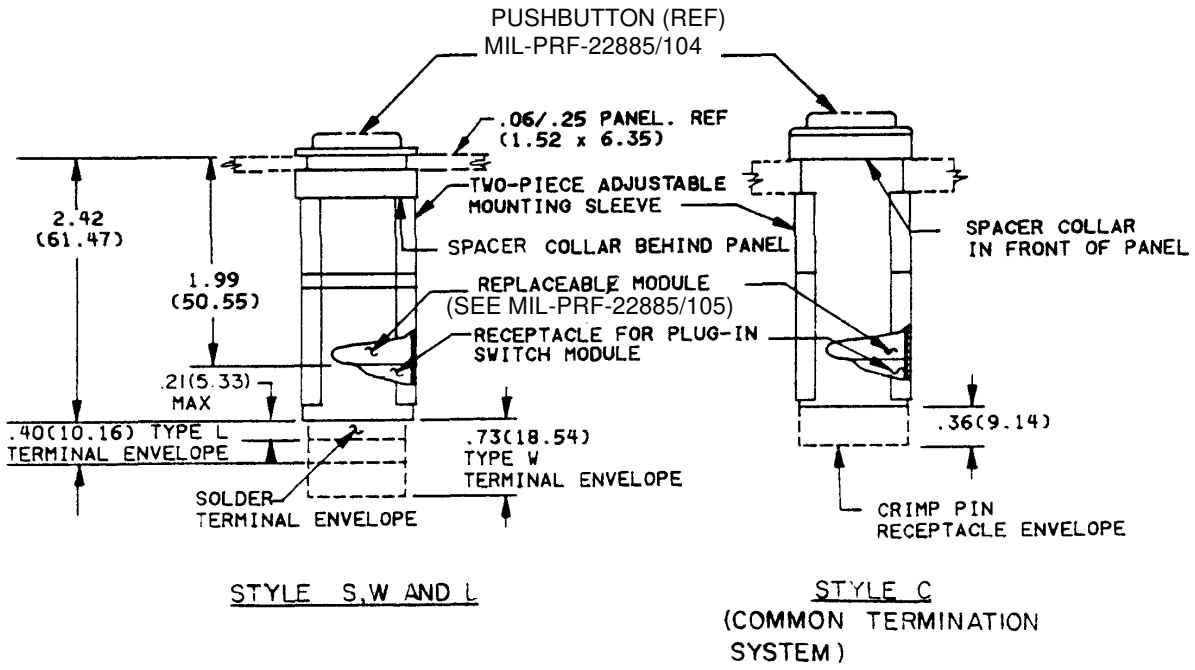
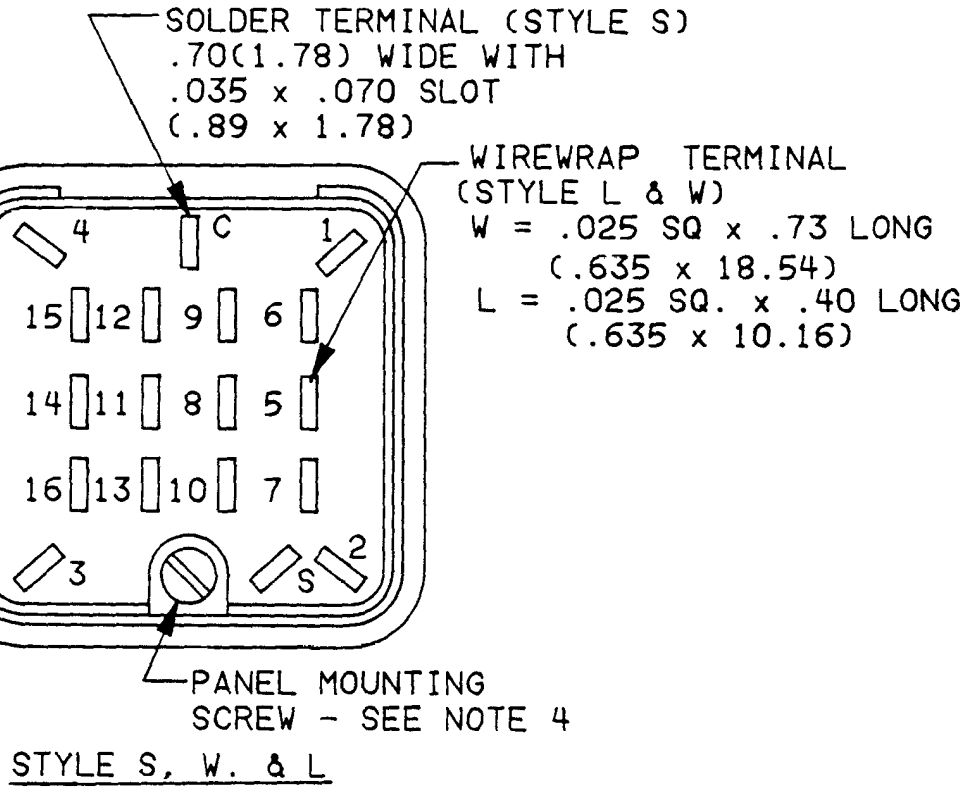
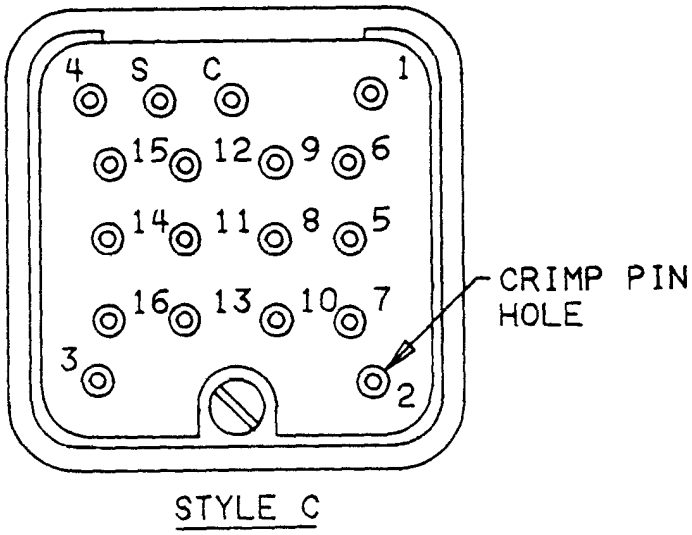


FIGURE 1. Push button switch assemblies.

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(COMMON TERMINATION SYSTEM)

FIGURE 1. Pushbutton switch assemblies - Continued.

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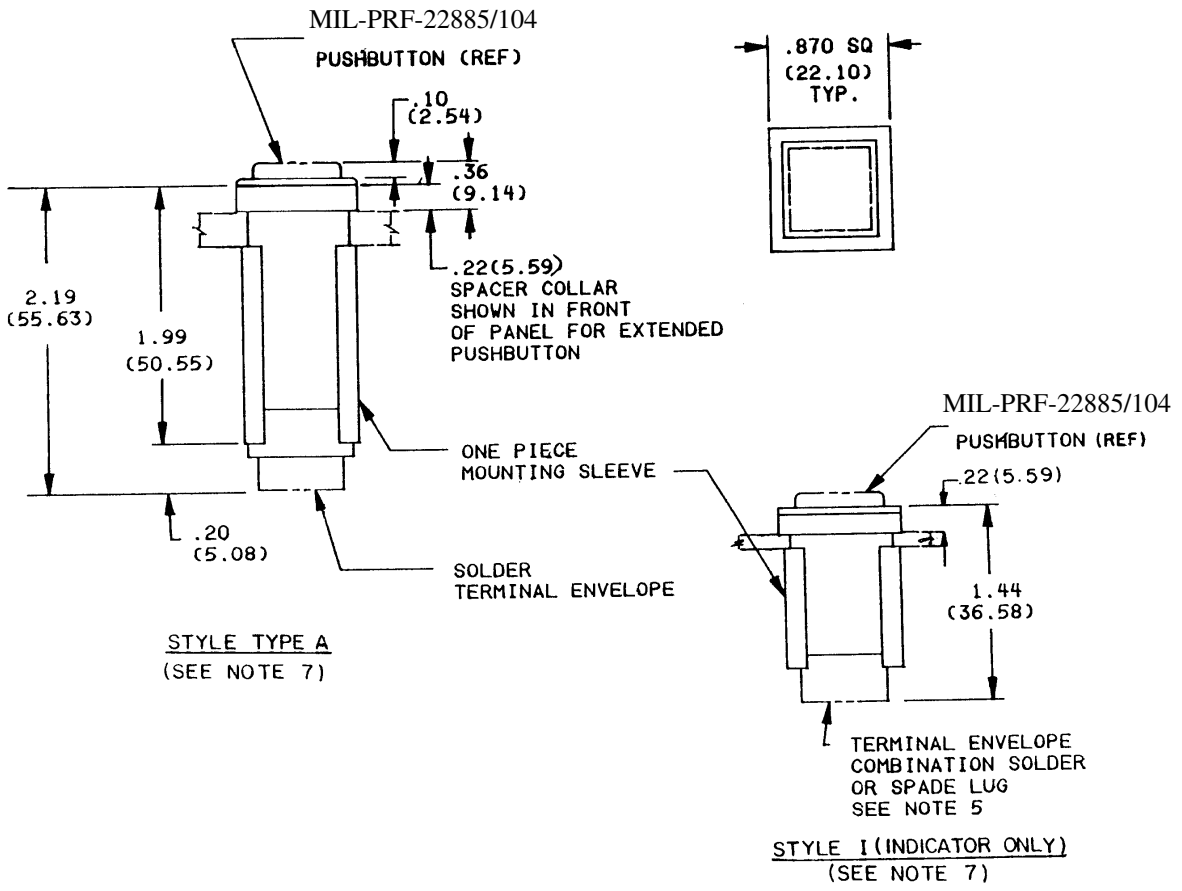
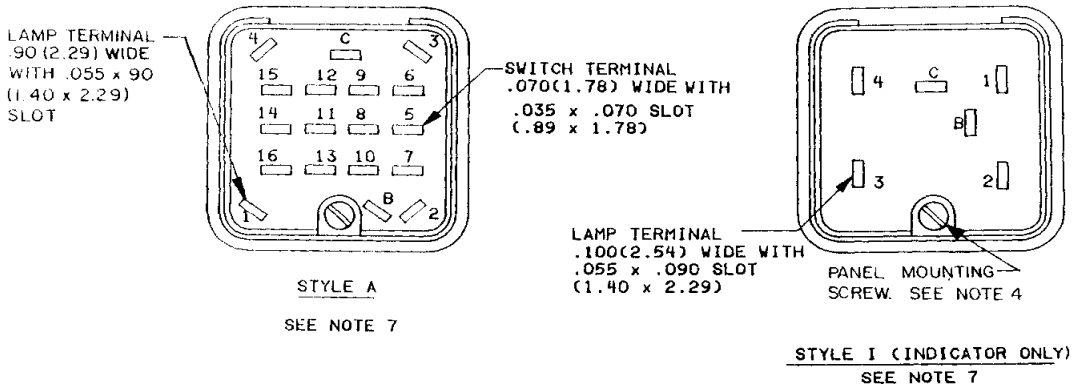


FIGURE 1. Pushbutton switch assemblied - Continued.

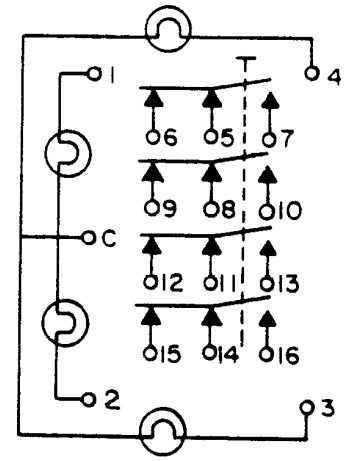
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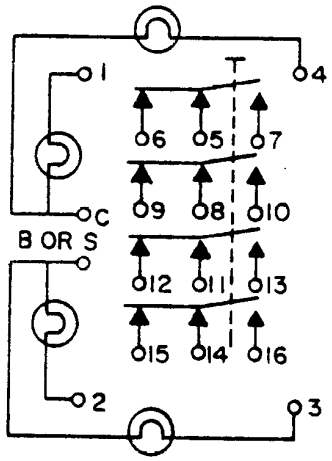
NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.015 (.38 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
3. Legend area for sunlight readable pushbutton display is .591 (15.01 mm) sq.
4. Maximum recommended tightening torque is 8 inch ounces on panel mounting screw.
5. Style I (indicator) terminals are a combination solder lug and spade lug. Each terminal will accept two 20 AWG stranded wires or a “fast-on” spade lug such as AMP #60537-1 or equivalent.
6. Unit is built with only the required terminals, see figure 2.
7. Style A and Style 1 are constructed with integral housing and switch/indicator, and do not accommodate MIL-PRF-22885/105 plug-in-modules.
8. Metric equivalents are in parentheses.

FIGURE 1. Pushbutton switch assemblies - Continued.

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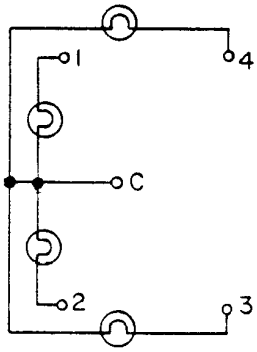


COMMON LAMP GROUND (CLG)

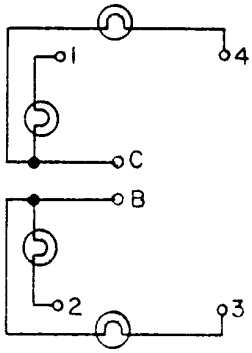


SPLIT LAMP GROUND (SLG)

SWITCH & LAMP SCHEMATICS, STYLE A, C, S, W & L



COMMON LAMP GROUND (CLG)



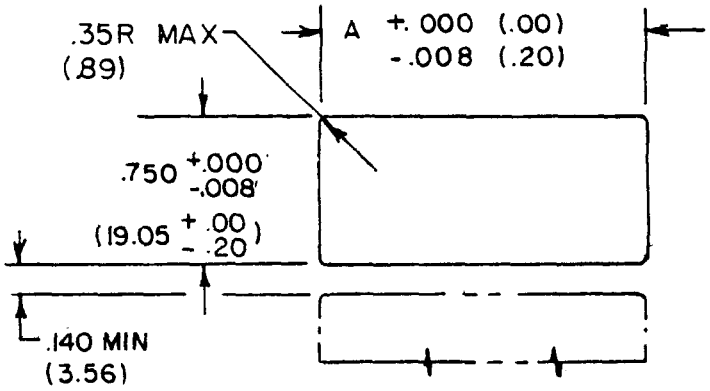
SPLIT LAMP GROUND (SLG)

LAMP SCHEMATICS, STYLE I INDICATOR ONLY

NOTES:
1. Delete pins 8 through 13 for 2PDT configurations.
2. On momentary operation switches, terminals 6, 9, 12 and 15 are normally closed.

FIGURE 2. Lamp and switch schematics.

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No. units	Dimension A
1	.750 (19.05)
2	1.621 (41.17)
3	2.492 (63.30)
4	3.363 (85.42)
N	A = .750 + .871 x N (19.05 + 22.12) x N

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.015 (.38 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.
3. Dimensions apply for both vertical and horizontal (shown) slot mounting.
4. Metric equivalents are in parentheses.

FIGURE 3. Horizontal and vertical mounting panel cutout dimensions.

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REQUIREMENTS:

Design and construction: See figures 1 and 2.

Complete switch shall consist of:

1- Switch/Indicator housing. Includes mounting sleeve, mounting screw with lockwasher, and spacer collar.

1- Replaceable switch/indicator module (styles S, W, L, and C only) in accordance with MIL-PRF-22885/105.

1- Pushbutton in accordance with MIL-PRF-22885/104. Accommodates 4 midget flanged base T-1 lamps. Pushbutton and lamps are acquired separately.

Classification:

Enclosure design: 1 (Unsealed).

Temperature characteristic: 1(-55°C to +85°C).

Vibration grade: 3(10 - 2,000 Hz, 15 g peak).

Operation:

A (Momentary action).

B (Alternate action).

H (Indicator only).

Color: See MIL-PRF-22885/104.

Display type:

S (Sunlight readable).

H (Hidden legend).

Material:

Replaceable switch/indicator module: See MIL-PRF-22885/105 (applicable to styles S, W, L, and C only).

Pushbutton: See MIL-PRF-22885/104.

Common termination system (applicable to style C switches):

Terminal junction system: Shall accept M39029/1-101 connector contacts. Shall accept MS27488-20 end seal plugs.

Terminal sealing grommet: Shall seal on smooth wire insulations of .038 (.97 mm) to .083 (2.11 mm) diameter.

Installation/removal tool: M81969/14-02.

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Weight (±10%):

Style	Ounces	Grams
I	.71	20
A	1.20	34
C	1.90	54
S, W, and L	1.90	54

Operating characteristics:

Actuation force: 3.5 ±1.0 pounds (1.59 ±.45 kg)
Release force: 0.4 (.18kg) pound minimum.
Actuator travel: .20 inch (5.08 mm) nominal.

Terminal strength:

Common termination system (style C): Using crimping tool M22520/1-01 and positioner M22520/1-02 to install connector contact on wire, and insertion/extraction tool M81969/14-02, the crimped wire contact shall be inserted and extracted 10 times. The insertion and extraction force shall not exceed 10 pounds. The contact shall again be inserted, and must hold axial load of 20 pounds minimum for 10 seconds.

Shock: Method I.

Electrical endurance: 25,000 cycles. See table I for test loads.

TABLE I. Electrical ratings.

Load	Sea level and 70,000 feet		Sea level
	28 V dc	115 V ac	30 mV 1/
Resistive Inductive Lamp	2 Amperes 1 Amperes .75 Ampere	2 Amperes 1 Amperes .75 Ampere	.01 Ampere

1/ Maximum dc or peak ac, 50,000 cycles. Low level life test conditions only.

Dielectric withstanding voltage:

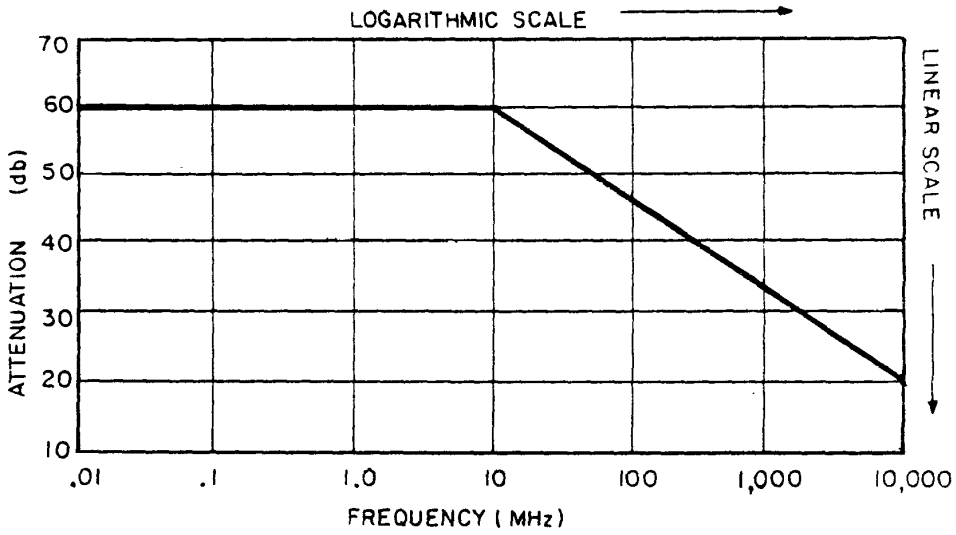
Sea level: 1,056 V rms.
70,000 ft: 450 V rms.

Logic level life: Applicable.

EMI/RFI shielding: Positive grounding and shielding efficiency test applicable when specified, see table III. The minimum shielding attenuation shall be as follows:

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
15 kHz	60 dB	100 MHz	47 dB
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB

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Illuminated color: See MIL-PRF-22885/104.

Luminance: See MIL-PRF-22885/104

Sunlight readability: Applicable except units are tested for $_1$ = 45° and $_2$ = 0° only. See MIL-PRF-22885/104 for contrast ratios.

Marking: Part numbers for complete assemblies shall not appear on the items. The part numbers for individual plug-in modules and housing shall be marked on the items as required.

Terminal identification: Terminal identification shall be shown on the housing terminal header or grommet.

Part number: M22885/103- (dash number, see table III). For acquisition of government spares for styles C, S, L and W, housing and switch module (per MIL-PRF-22885/105) shall be acquired separately.

Qualification inspection:

Group submission: Table II applies provided that similar products are qualified to MIL-PRF-22885/56 and MIL-PRF-22885/85.

Quality conformance inspection:

Group B inspection:

EMI/RFI shielding: Only positive grounding test need be performed.

Group A inspection: Do not exceed 6 V dc or peak ac, 10 milliamperes during group A inspection.

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TABLE II. Qualification inspection (group inspection).

Qualification inspection table of MIL-PRF-22885			Additional inspections	Extent of approval
Switch	Group	Number of samples		
M22885/103-044 3/ 5/	I II III 1/ VI 2/ VII	All samples 4 2 2 2	7/ 8/	All
M22885/103-031 4/ 5/	I II 1/ VIII	All samples 4 2 each color 6/		

1/ These inspections may be omitted, provided that similar products are qualified to MIL-PRF-22885/95.
2/ Resistive dc loads only (sea level and altitude).
3/ Units shall be equipped with M22885/104-10 pushbuttons and 4 MS24515-718 lamps.
4/ Units shall be equipped with M22885/104-7 pushbuttons and 4 MS24515-718 lamps.
5/ Legend shall consist of 2 lines of characters which utilize at least ¾ of the maximum legend area.
6/ Two pushbuttons of each color may be installed on housings submitted to the inspections of group II.
7/ Two sample units each of styles S, W, L, A and I assemblies shall be inspected for visual and mechanical examination and operating characteristics.
8/ Two sample units each of style I indicator assemblies shall be inspected for solderability, resistance to soldering heat, and lamp contact resistance.

TABLE III Part numbers.

Part number M22885/103	Style	Termination	Operation	Switch circuit	Lamp circuit	EMI/RFI shielding
001	I	Solder	Indicator	N/A	CLG	No
002	I	Solder	Indicator	N/A	SLG	No
003	I	Solder	Indicator	N/A	CLG	Yes
004	I	Solder	Indicator	N/A	SLG	Yes
005	A	Solder	Indicator	N/A	CLG	No
006	A	Solder	Indicator	N/A	SLG	No
007	A	Solder	Indicator	N/A	CLG	Yes
008	A	Solder	Indicator	N/A	SLG	Yes
009	A	Solder	Momentary	4PDT	CLG	No
010	A	Solder	Momentary	4PDT	SLG	No
011	A	Solder	Momentary	2PDT	CLG	No
012	A	Solder	Momentary	2PDT	SLG	No
013	A	Solder	Alternate	4PDT	CLG	No
014	A	Solder	Alternate	4PDT	SLG	No
015	A	Solder	Alternate	2PDT	CLG	No
016	A	Solder	Alternate	2PDT	SLG	No
017	A	Solder	Momentary	4PDT	CLG	Yes
018	A	Solder	Momentary	4PDT	SLG	Yes
019	A	Solder	Momentary	2PDT	CLG	Yes
020	A	Solder	Momentary	2PDT	SLG	Yes
021	A	Solder	Alternate	4PDT	CLG	Yes
022	A	Solder	Alternate	4PDT	SLG	Yes
023	A	Solder	Alternate	2PDT	CLG	Yes
024	A	Solder	Alternate	2PDT	SLG	Yes
025	C	Crimp pin	No module	4PDT	SLG	No

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TABLE III Part numbers - Continued.

Part number M22885/103	Style	Termination	Operation	Switch circuit	Lamp circuit	EMI/RFI shielding
026	C	Crimp pin	No module	4PDT	SLG	Yes
027	C	Crimp pin	Indicator	N/A	CLG	No
028	C	Crimp pin	Indicator	N/A	SLG	No
029	C	Crimp pin	Indicator	N/A	CLG	Yes
030	C	Crimp pin	Indicator	N/A	SLG	Yes
031	C	Crimp pin	Momentary	4PDT	CLG	No
032	C	Crimp pin	Momentary	4PDT	SLG	No
033	C	Crimp pin	Momentary	2PDT	CLG	No
034	C	Crimp pin	Momentary	2PDT	SLG	No
035	C	Crimp pin	Alternate	4PDT	CLG	No
036	C	Crimp pin	Alternate	4PDT	SLG	No
037	C	Crimp pin	Alternate	2PDT	CLG	No
038	C	Crimp pin	Alternate	2PDT	SLG	No
039	C	Crimp pin	Momentary	4PDT	CLG	Yes
040	C	Crimp pin	Momentary	4PDT	SLG	Yes
041	C	Crimp pin	Momentary	2PDT	CLG	Yes
042	C	Crimp pin	Momentary	2PDT	SLG	Yes
043	C	Crimp pin	Alternate	4PDT	CLG	Yes
044	C	Crimp pin	Alternate	4PDT	SLG	Yes
045	C	Crimp pin	Alternate	2PDT	CLG	Yes
046	C	Crimp pin	Alternate	2PDT	SLG	Yes
047	S	Solder	No module	4PDT	SLG	No
048	S	Solder	No module	4PDT	SLG	Yes
049	S	Solder	Indicator	N/A	CLG	No
050	S	Solder	Indicator	N/A	SLG	No
051	S	Solder	Indicator	N/A	CLG	Yes
052	S	Solder	Indicator	N/A	SLG	Yes
053	S	Solder	Momentary	4PDT	CLG	No
054	S	Solder	Momentary	4PDT	SLG	No
055	S	Solder	Momentary	2PDT	CLG	No
056	S	Solder	Momentary	2PDT	SLG	No
057	S	Solder	Alternate	4PDT	CLG	No
058	S	Solder	Alternate	4PDT	SLG	No
059	S	Solder	Alternate	2PDT	CLG	No
060	S	Solder	Alternate	2PDT	SLG	No
061	S	Solder	Momentary	4PDT	CLG	Yes
062	S	Solder	Momentary	4PDT	SLG	Yes
063	S	Solder	Momentary	2PDT	CLG	Yes
064	S	Solder	Momentary	2PDT	SLG	Yes
065	S	Solder	Alternate	4PDT	CLG	Yes
066	S	Solder	Alternate	4PDT	SLG	Yes
067	S	Solder	Alternate	2PDT	CLG	Yes
068	S	Solder	Alternate	2PDT	SLG	Yes
069	L	.40 wire wrap	No module	4PDT	SLG	No
070	L	.40 wire wrap	No module	4PDT	SLG	Yes
071	L	.40 wire wrap	Indicator	N/A	CLG	No
072	L	.40 wire wrap	Indicator	N/A	SLG	No
073	L	.40 wire wrap	Indicator	N/A	CLG	Yes
074	L	.40 wire wrap	Indicator	N/A	SLG	Yes
075	L	.40 wire wrap	Momentary	4PDT	CLG	No

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TABLE III Part numbers - Continued.

Part number M22885/103	Style	Termination	Operation	Switch circuit	Lamp circuit	EMI/RFI shielding
076	L	.40 wire wrap	Momentary	4PDT	SLG	No
077	L	.40 wire wrap	Momentary	2PDT	CLG	No
078	L	.40 wire wrap	Momentary	2PDT	SLG	No
079	L	.40 wire wrap	Alternate	4PDT	CLG	No
080	L	.40 wire wrap	Alternate	4PDT	SLG	No
081	L	.40 wire wrap	Alternate	2PDT	CLG	No
082	L	.40 wire wrap	Alternate	2PDT	SLG	No
083	L	.40 wire wrap	Momentary	4PDT	CLG	Yes
084	L	.40 wire wrap	Momentary	4PDT	SLG	Yes
085	L	.40 wire wrap	Momentary	2PDT	CLG	Yes
086	L	.40 wire wrap	Momentary	2PDT	SLG	Yes
087	L	.40 wire wrap	Alternate	4PDT	CLG	Yes
088	L	.40 wire wrap	Alternate	4PDT	SLG	Yes
089	L	.40 wire wrap	Alternate	2PDT	CLG	Yes
090	L	.40 wire wrap	Alternate	2PDT	SLG	Yes
091	W	.73wire wrap	No module	4PDT	SLG	No
092	W	.73wire wrap	No module	4PDT	SLG	Yes
093	W	.73wire wrap	Indicator	N/A	CLG	No
094	W	.73wire wrap	Indicator	N/A	SLG	No
095	W	.73wire wrap	Indicator	N/A	CLG	Yes
096	W	.73wire wrap	Indicator	N/A	SLG	Yes
097	W	.73wire wrap	Momentary	4PDT	CLG	No
098	W	.73wire wrap	Momentary	4PDT	SLG	No
099	W	.73wire wrap	Momentary	2PDT	SLG	No
100	W	.73wire wrap	Momentary	2PDT	CLG	No
101	W	.73wire wrap	Alternate	4PDT	SLG	No
102	W	.73wire wrap	Alternate	4PDT	CLG	No
103	W	.73wire wrap	Alternate	2PDT	SLG	No
104	W	.73wire wrap	Alternate	2PDT	CLG	No
105	W	.73wire wrap	Momentary	4PDT	SLG	Yes
106	W	.73wire wrap	Momentary	4PDT	CLG	Yes
107	W	.73wire wrap	Momentary	2PDT	SLG	Yes
108	W	.73wire wrap	Momentary	2PDT	CLG	Yes
109	W	.73wire wrap	Alternate	4PDT	SLG	Yes
110	W	.73wire wrap	Alternate	4PDT	CLG	Yes
111	W	.73wire wrap	Alternate	2PDT	SLG	Yes
112	W	.73wire wrap	Alternate	2PDT	CLG	Yes

Custodians:
Navy - EC

Preparing activity:
DLA - CC

(Project 5930-N779-01)

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